Jr.Pen 5.1. Walking An Application

Task 1 - Walking An Application

Here is a short breakdown of the in-built browser tools:

- **View Source** - Use your browser to view the human-readable source code of a website.
- **Inspector** - Learn how to inspect page elements and make changes to view usually blocked content.
- **Debugger** - Inspect and control the flow of a page's JavaScript
- **Network** - See all the network requests a page makes.

Task 2 - Exploring The Website

As a penetration tester, your role when reviewing a website or web application is to discover features that could potentially be vulnerable and attempt to exploit them to assess whether or not they are. These features are usually parts of the website that require some interactivity with the user. Finding interactive portions of the website can be as easy as spotting a login form to manually reviewing the website's JavaScript. An excellent place to start is just with your browser exploring the website and noting down the individual pages/areas/features with a summary for each one.

Task 3 - Viewing The Page Source

The page source is the human-readable code returned to our browser/client from the web server each time we make a request. The returned code is made up of HTML (HyperText Markup Language), CSS (Cascading Style Sheets) and JavaScript, and it's what tells our browser what content to display, how to show it and adds an element of interactivity with JavaScript.

```
            </div>
            <div id="navbar" class="collapse navbar-collapse">
                <ul class="nav navbar-nav">
                    <li class="active"><a href="/">Home</a></li>
                    <li><a href="/news">News</a></li>
                    <li><a href="/contact">Contact</a></li>
                    <li><a href="/customers">Customers</a></li>
                </ul>
            </div><!--/.nav-collapse -->
```

Task 4 - Developer Tools - Inspector

Developer Tools: Every modern browser includes developer tools; this is a tool kit used to aid web developers in debugging web applications and gives you a peek under the hood of a website to see what is going on. As a pentester, we can leverage these tools to provide us with a much better understanding of the web application. We're specifically focusing on three features of the developer tool kit, Inspector, Debugger and Network.

Opening Developer Tools: The way to access developer tools is different for every browser. If you're not sure how to access it, click the "View Site" button on the top right of this task to get instructions to how to access the tools for your browser.

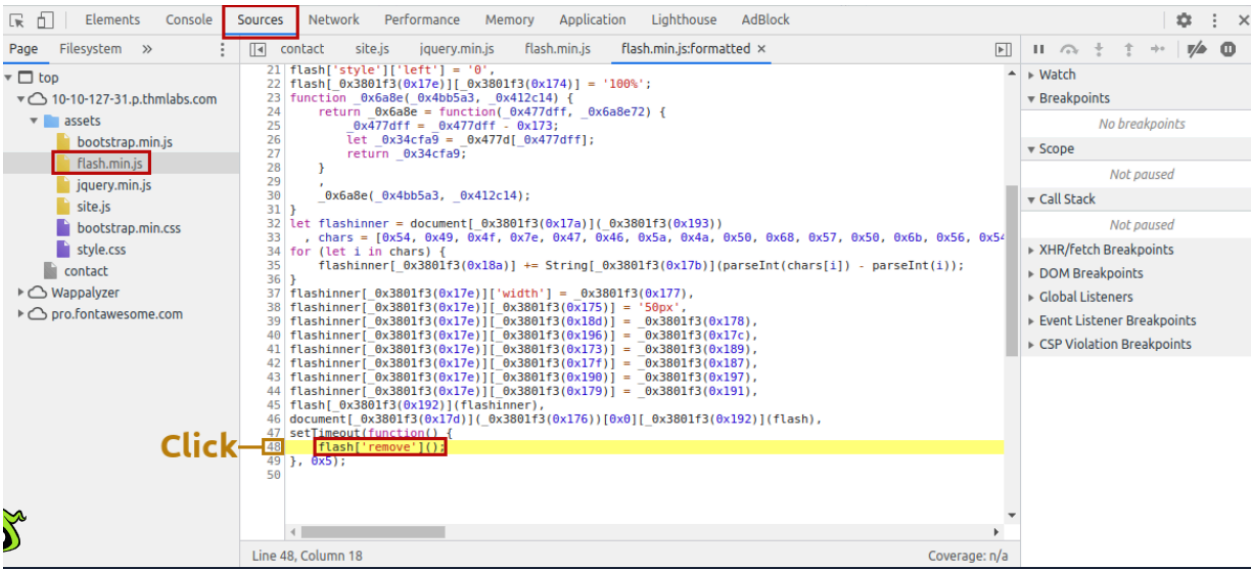
Inspector: The page source doesn't always represent what's shown on a webpage; this is because CSS, JavaScript and user interaction can change the content and style of the page, which means we need a way to view what's been displayed in the browser window at this exact time. Element inspector assists us with this by providing us with a live representation of what is currently on the website.

As well as viewing this live view, we can also edit and interact with the page elements, which is helpful for web developers to debug issues. On the Acme IT Support website, click into the news section, where you'll see three news articles.

The first two articles are readable, but the third has been blocked with a floating notice above the content stating you have to be a premium customer to view the article. These floating boxes blocking the page contents are often referred to as paywalls as they put up a metaphorical wall in front of the content you wish to see until you pay.

Task 5 - Developer Tools - Debugger

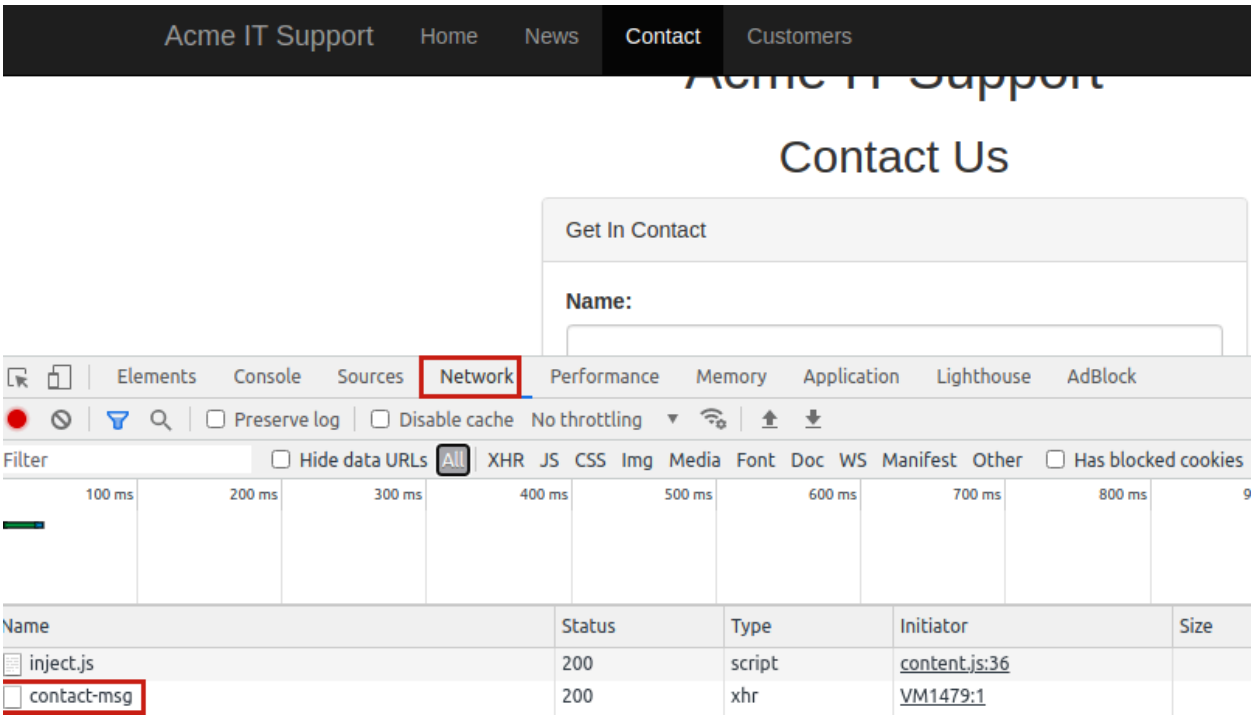
Developer Tools - Debugger: This panel in the developer tools is intended for debugging JavaScript, and again is an excellent feature for web developers wanting to work out why something might not be working. But as penetration testers, it gives us the option of digging deep into the JavaScript code. In Firefox and Safari, this feature is called Debugger, but in Google Chrome, it's called Sources. Many times when viewing javascript files, you'll notice that everything is on one line, which is because it has been minimised, which means all formatting (tabs, spacing and newlines) have been removed to make the file smaller. This file is no exception to this, and it has also been obfusticated, which makes it purposely difficult to read, so it can't be copied as easily by other developers.



Task 6 - Developer Tools - Network

Developer Tools - Network: The network tab on the developer tools can be used to keep track of every external request a webpage makes. If you click on the Network tab and then refresh the page, you'll see all the files the page is requesting. Try doing this on the contact page; you can press the trash can icon to delete the list if it gets a bit overpopulated.

With the network tab open, try filling in the contact form and pressing the **Send Message** button. You'll notice an event in the network tab, and this is the form being submitted in the background using a method called AJAX. AJAX is a method for sending and receiving network data in a web application background without interfering by changing the current web page.


Acme IT Support
Home
News
Contact
Customers
Contact Us
Get In Contact
Name:
Elements
Console
Sources
Network
Performance
Memory
Application
Lighthouse
AdBlock
Preserve log
Disable cache
No throttling
Filter
Hide data URLs
All
XHR
JS
CSS
Img
Media
Font
Doc
WS
Manifest
Other
Has blocked cookies
100 ms
200 ms
300 ms
400 ms
500 ms
600 ms
700 ms
800 ms
Name
Status
Type
Initiator
Size
inject.js
200
script
content.js:36
contact-msg
200
xhr
VM1479:1